

Customer Churn Analytics & AI Prediction

Data Analytics & AI Internship Project

1. Abstract

This project combines data analytics and artificial intelligence to analyze customer churn and predict whether a customer may leave a service. The workflow includes data preparation, exploratory analysis, visualization, machine learning and evaluation.

2. Objectives

Prepare customer data, identify churn patterns, visualize trends, train an AI classification model, evaluate its performance and identify useful predictive features.

3. Dataset

A reproducible synthetic dataset containing 1,200 customer records was generated. It includes age, tenure, monthly charges, support calls, satisfaction, contract, internet service, payment method, technical support and churn status.

4. Preprocessing

Missing numeric values are filled with the median, categorical values with the most frequent category, categorical variables are one-hot encoded and numeric variables are standardized using a scikit-learn pipeline.

5. Exploratory Analysis

The project visualizes overall churn distribution, churn rate by contract type and churn rate versus satisfaction score.

6. AI Model

A Random Forest Classifier with 250 trees is trained using an 80/20 stratified train-test split. Balanced class weights are used to account for the two churn classes.

7. Evaluation

Accuracy, precision, recall and F1 score are calculated. A confusion matrix is also generated.

8. Business Insight

In this demonstration, factors such as contract type, tenure, satisfaction, support interactions and service characteristics provide useful signals for churn-risk analysis. Feature importance is included in the output.

9. Conclusion

The project demonstrates a complete Data Analytics and AI workflow. A real deployment would require validated organizational data, privacy controls, monitoring and regular retraining.

10. Model Results

Metric	Value
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Accuracy	0.733
Precision	0.606
Recall	0.513
F1 Score	0.556

Figure 1: Churn distribution

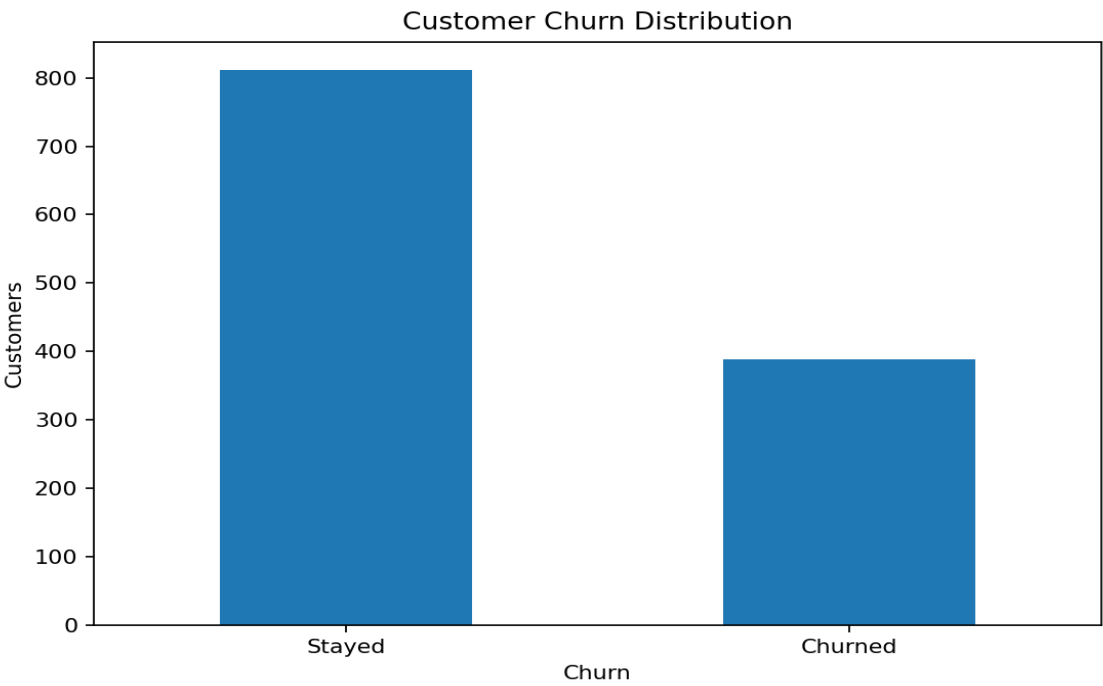


Figure 2: Churn rate by contract type

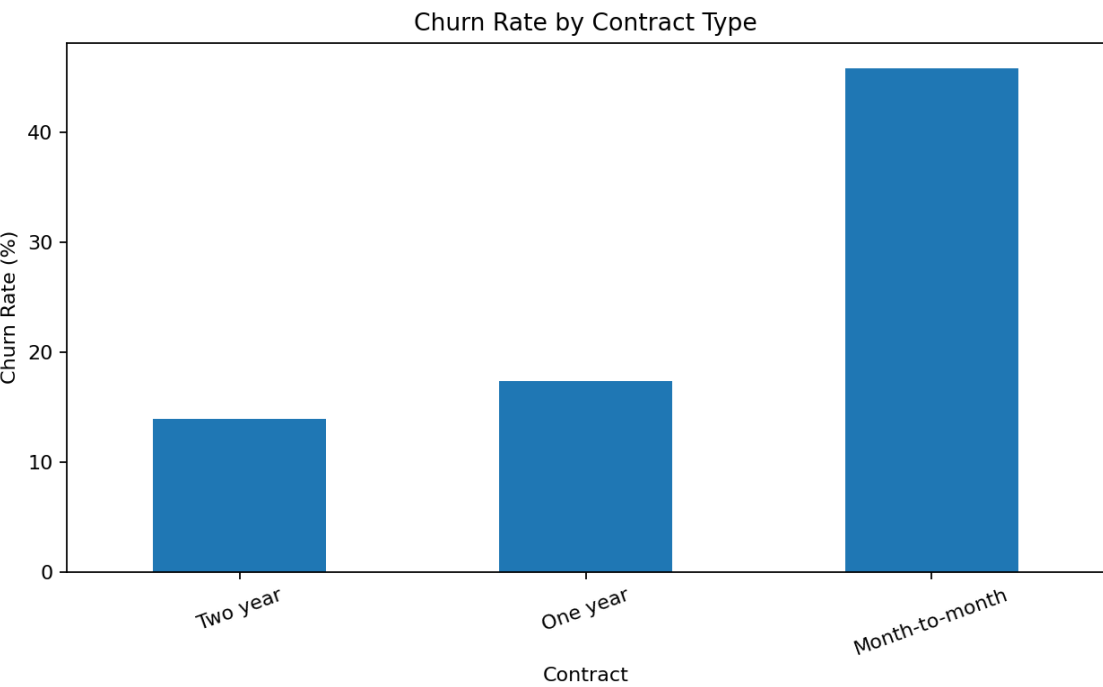


Figure 3: Churn rate versus satisfaction

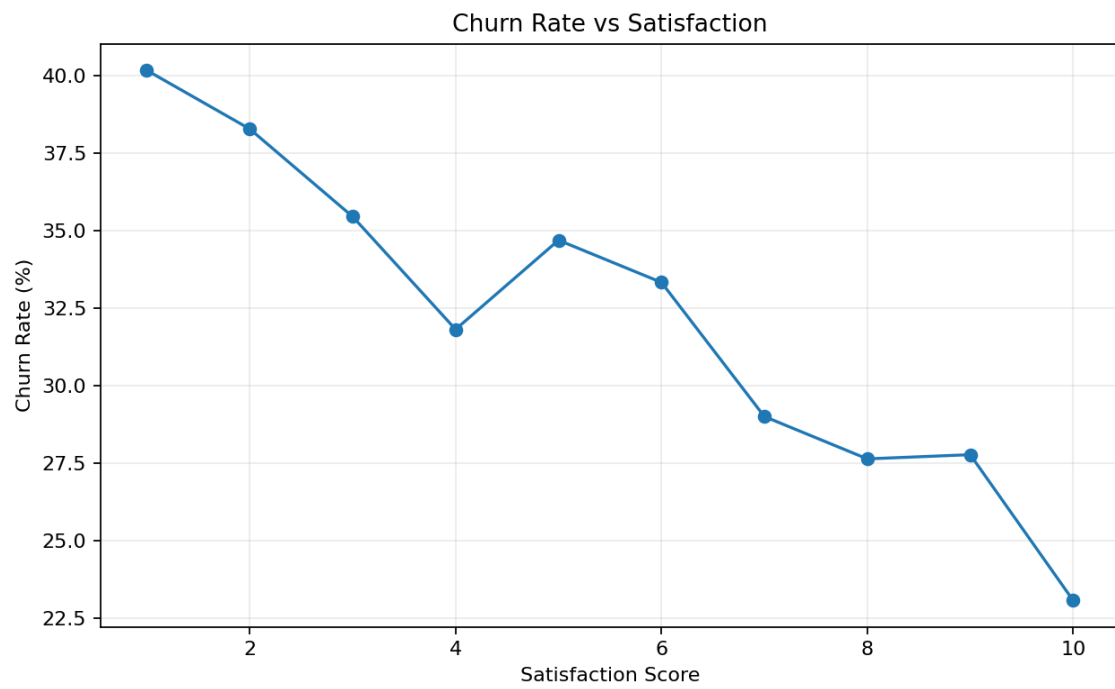


Figure 4: Top predictive features

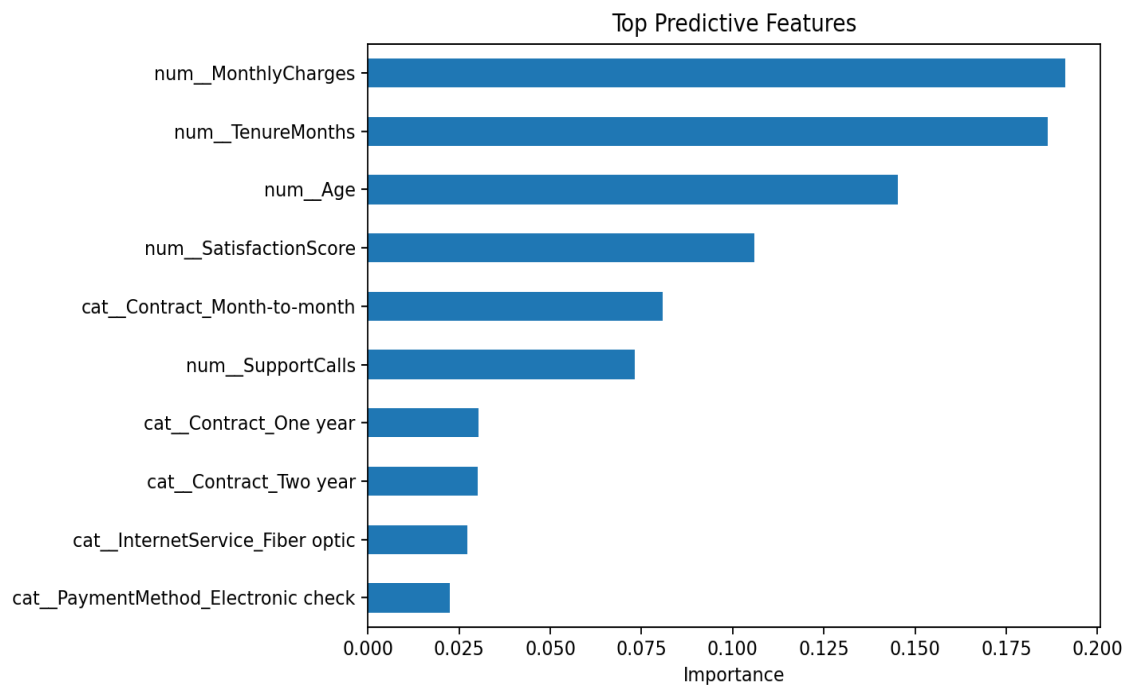
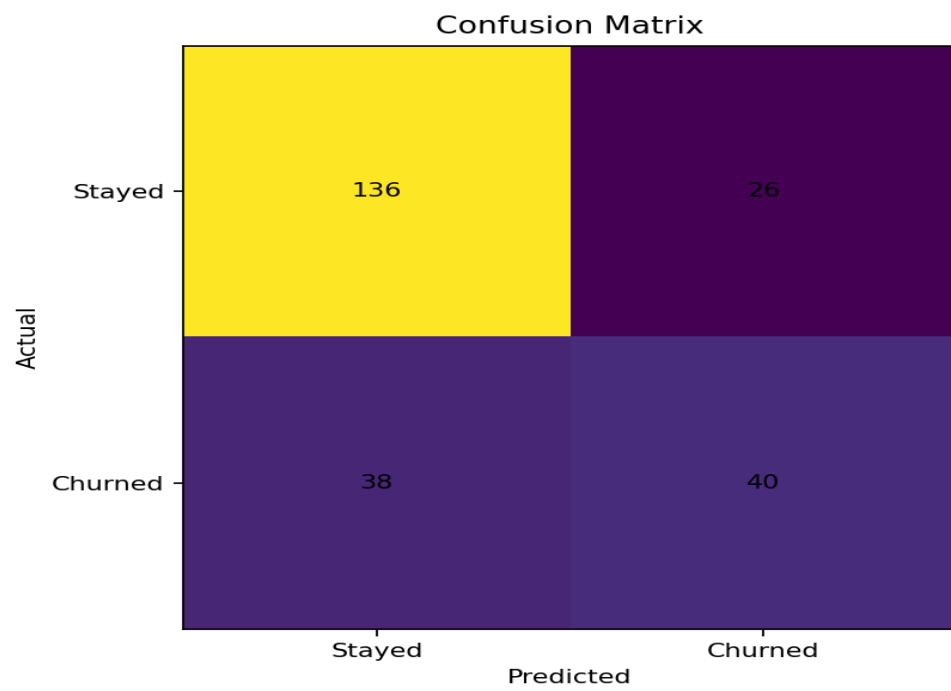


Figure 5: Confusion matrix



Note: This internship demonstration uses synthetic data. Results are for demonstration and are not production performance claims.